The chain

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Chapter 05 Stage 5 of 14 Extreme concentration · Hard to substitute

Photomasks and Pellicles

Only three firms in the world make the blank plates for the masks used to print the finest layers of leading-edge chips. Japan's AGC and Hoya are two of them.

Specimen 05 · Photomasks and Pellicles

An EUV photomask, or reticle, with its pellicle dust cover. The glass plate is drawn to scale. It is a 152 mm square, 6.35 mm thick, about the size of a small floor tile. The dark blue band around the top edge is the mirror coating. The real coating is about forty pairs of layers, each pair 7 nm thick, far thinner than drawn.

In plain terms

A photomask is the stencil that lithography prints from, at least one for each layer of a chip[1]. The same mask prints its layer on wafer after wafer, so one mask shapes millions of chips. The most complicated layers are printed with extreme ultraviolet (EUV) light, and the rest can often be printed with deep ultraviolet light[2]. For deep ultraviolet the mask is a quartz plate that the light shines through. No glass lets EUV light through, so the mask for EUV layers is a mirror. The mirror is built up from about forty pairs of thin layers, and the light bounces off it. Japan's AGC and Hoya are two of the three firms that make these mirrors. A dark pattern drawn on top of the mirror soaks up the light wherever no line should print. Any speck of dust on the mask repeats on every wafer it prints. So a thin membrane on a frame, called a pellicle, sits just above the mask and catches dust where it is too far out of focus to print.

In short

On CSET's count from TechInsights data, Austria's IMS Nanofabrication held 46.6 percent of the 2025 world market for electron-beam mask writers, which draw the pattern onto a mask[3]. Hoya says its share of the mask blank market is exceptionally high[4].

1 PHOTRONICS INC, Form 10-K annual report for the period ended 2025-10-31 (10-K). U.S. Securities and Exchange Commission, 17 December 2025. sec.gov

2 See ASML's DUV lithography systems. ASML, accessed 9 September 2026. asml.com

3 Supply Chain Explorer: Advanced Chips. Emerging Technology Observatory (ETO), accessed 9 September 2026. chipexplorer.eto.tech

4 HOYA, Information Technology business, review of operations, Integrated Report... HOYA, accessed 9 September 2026. hoya.com

SemiAnalysis put a full set of masks for a leading-edge chip at $40 million in May 2026[5]. In July 2022 it wrote that more companies will lack the volume to spread the fixed cost of mask sets[6]. On 8 September 2026 ASML and TSMC announced a plan to move from 6-inch to 12-inch masks[7]. They expect the change to raise fab productivity and lower chipmaking costs[7].

Each EUV layer of an AI accelerator is printed from a mask built on a blank plate. Only three companies worldwide supply those plates, and Japan's AGC is one of them[8]. Hoya, also Japanese, is another[4]. AGC says it is the only company in the world that takes an EUV blank all the way from the glass to the finished coating[9].

Chokepoint card Photomasks and Pellicles, stage 5 of 14

Concentration

Extreme

Three or fewer firms could supply this stage today, or one firm holds more than half of it.

Substitutability

Hard

AGC said in December 2024 that only three firms supply the blank plates for EUV masks[8].

Price or market size

$40M

for a leading-edge mask set, SemiAnalysis's May 2026 figure[5]. In July 2022 it put a 7 nm set above $10M[6]

Who leads	JP	**AGC**	Ranks itself No.2 worldwide in EUV mask blanks on its own fiscal 2026 estimates[10]
	JP	**Hoya**	Said in its 2026 report that its share of mask blanks is exceptionally high[4]
	JP	**Lasertec**	Launched the world's first tool that inspects patterned EUV masks at the scanner's own wavelength in 2019[11]
	AT	**IMS Nanofabrication**	CSET puts its 2025 share of the market for all electron-beam mask writers at 46.6%, using TechInsights data[3]
	US	**Photronics**	$849.3M of revenue in fiscal 2025[12]

3 Supply Chain Explorer: Advanced Chips. Emerging Technology Observatory (ETO), accessed 9 September 2026. chipexplorer.eto.tech

4 HOYA, Information Technology business, review of operations, Integrated Report... HOYA, accessed 9 September 2026. hoya.com

5 EDA Market Primer. SemiAnalysis, 21 May 2026. newsletter.semianalysis.com

6 The Dark Side Of The Semiconductor Design Renaissance – Fixed Costs Soaring Due... SemiAnalysis, 24 July 2022. newsletter.semianalysis.com

7 ASML and TSMC Announce Initiative to Pioneer Industry Transition to... TSMC, 8 September 2026. pr.tsmc.com

8 The Japanese glass behind next-generation chips. AGC, 13 December 2024. agc.com

9 AGC to Boost Production Capacity of EUVL Photomask Blanks. AGC, 27 April 2023. agc.com

10 AGC Data Book. AGC Inc., 29 May 2026, p. 8. agc.com

11 Lasertec Releases ACTIS A200HiT Series Actinic EUV Patterned Mask Inspection... Lasertec, 31 October 2025. lasertec.co.jp

12 Photronics Reports Full Year and Fourth Quarter Fiscal 2025 Results. GlobeNewswire (release by Photronics), 10 December 2025. globenewswire.com

Where it is made	JP	**Japan**	Mask blanks (AGC, Hoya), DUV blanks (Shin-Etsu), inspection (Lasertec), writers (NuFlare), pellicles (Mitsui Chemicals)
	AT	**Austria**	IMS Nanofabrication multi-beam mask writers
	DE	**Germany**	Zeiss mask checking in EUV light (AIMS EUV) and mask repair
	US	**United States**	Photronics merchant mask shops, KLA inspection
	TW	**Taiwan**	TSMC, which held 10% of IMS at the end of 2024

Why substitution is slow

AGC says it spent more than a decade meeting the EUV blank's requirements, such as near-zero thermal expansion, and began mass production only in 2017[8]. Germany's Zeiss calls its AIMS the only system that tests how a mask will print under conditions equivalent to the scanner's[13].

Where China stands

As of 2026 SMIC runs two mask shops of its own in Shanghai[14]. No Chinese firm appears in CSET's 2025 market shares for electron-beam mask writers, the machines that draw the pattern onto a mask[3].

Where the US stands

In December 2025 Photronics called itself a leading worldwide maker of photomasks[12]. As of 2026 KLA sells tools that inspect EUV masks and optical masks, the quartz masks used with deep ultraviolet light[15].

01

How it works

A mask starts as a blank, a polished plate with no pattern on it yet. For deep ultraviolet the blank is a window of quartz carrying a light-blocking film such as molybdenum silicide[16]. Japan's Shin-Etsu is one maker of those blanks[16]. For EUV the blank is a mirror of about 40 molybdenum and silicon layer pairs, 7 nm to a pair, laid on glass[17]. A dark tantalum film on top, called the absorber, will carry the pattern[18].

The blank is coated with a sensitive film called resist. A mask writer then draws the design into the resist with electrons. A single-beam writer flashes one shape at a time, while NuFlare's multi-beam writer for the most advanced masks controls 260,000 beams at once[19].

A wash removes the drawn resist. An etch then removes the blocking film or absorber wherever the resist is gone, and the rest of the resist is stripped. The plate now passes light, or reflects it, in the shape of one layer of the chip. On standard EUV scanners the pattern is drawn

3 Supply Chain Explorer: Advanced Chips. Emerging Technology Observatory (ETO), accessed 9 September 2026. chipexplorer.eto.tech

8 The Japanese glass behind next-generation chips. AGC, 13 December 2024. agc.com

12 Photronics Reports Full Year and Fourth Quarter Fiscal 2025 Results. GlobeNewswire (release by Photronics), 10 December 2025. globenewswire.com

13 Mask Qualification Solutions by ZEISS SMT. Carl Zeiss, accessed 9 September 2026. zeiss.com

14 Mask Service. SMIC, accessed 9 September 2026. smics.com

15 Reticle Qualification. KLA Corporation, accessed 20 October 2026. kla.com

16 Photomask blanks. Shin-Etsu Chemical, 3 April 2019. shinetsu.co.jp

17 The refined EUV mask model. ArXiv, 19 December 2019, p. 2. arxiv.org

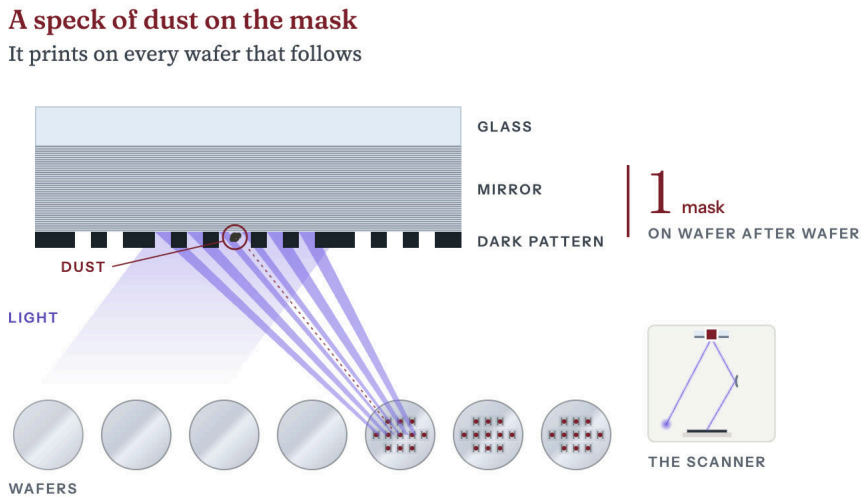
18 The refined EUV mask model. ArXiv, 19 December 2019, p. 1. arxiv.org

19 EB Mask Writer. NuFlare Technology, accessed 9 September 2026. nuflare.co.jp

four times larger than it will print[20]. An inspection tool compares the plate to the design, and a repair tool fixes the stray defects it finds.

Figure 05.1 · Explainer, 23 seconds

How a photomask prints its pattern



1. **The scanner.** A photomask is the stencil that lithography prints from, at least one for each layer of a chip. Inside the scanner, light bounces off the mask and carries its pattern down to the wafer.
2. **A mirror.** No glass lets extreme ultraviolet (EUV) light through. So the mask for EUV layers is a mirror, built up from about forty pairs of very thin layers laid on glass, and the light bounces off it.
3. **The pattern.** A dark pattern drawn on the mirror soaks up the light wherever no line should print. The light that bounces back carries the pattern of one layer.
4. **Wafer after wafer.** The same mask prints its layer on wafer after wafer, so one plate shapes millions of chips.
5. **One speck.** A speck of dust on the mask prints on every wafer, until someone finds it. So a thin membrane on a frame, called a pellicle, covers the mask's patterned side and catches dust where it is too far out of focus to print.

Note Simplified. The mask is drawn face down, as it sits in the scanner. The mirror layers and the dark pattern are drawn far thicker than they are, and the light's path through the scanner's mirrors is cut to one bounce.

EUV masks bring problems of their own:

- **Mask 3D effects.** Light strikes the mirror at a slant, and the absorber has height, so its edges cast small shadows that distort the printed shape. Software has to correct for the shadows in advance.
- **Write precision.** Every edge of the absorber has to sit within a few nanometers of where the design says.
- **Particles.** Anything on the mirror prints on every wafer until someone finds it.

[20] 5 things you should know about High NA in EUV. ASML, 25 January 2024. asml.com

High-NA scanners, the newest EUV machines, print finer lines with larger mirrors. To keep masks at their usual size, the scanners' mirrors shrink the pattern four times in one direction and eight times in the other[20]. So each exposure covers half the area of a standard exposure, and a wafer takes twice as many exposures[20].

On 8 September 2026 ASML and TSMC announced a plan to move from the current 6-inch masks to 12-inch masks[7]. They expect the change to raise fab productivity, lower chipmaking costs and remove stitching constraints, the design limits that come from printing a large chip in two halves[7]. They aim to set up a 12-inch mask pilot line by 2031, so that High-NA scanners using 12-inch masks can enter advanced chip production by 2033[7].

02

Variants and trade-offs

Blanks

AGC says it spent more than a decade meeting the blank's requirements, which include near-zero thermal expansion, perfectly smooth and flat surfaces and no particle contamination[8]. It says it began mass production only in 2017[8].

AGC publishes no market share figure for EUV blanks. In its data book, AGC ranks itself No.2 worldwide in EUV blanks, based on its own fiscal 2026 estimates[10]. Hoya says its share of the mask blank market is exceptionally high, and it calls itself the primary supplier for leading-edge EUV nodes[4]. It also says customers may gradually start buying EUV blanks from more than one supplier[4].

AGC planned to add EUV blank capacity by 2025, and it says its sales reached their target a year early. In April 2023 AGC said its EUV blank capacity would grow by about 30 percent in 2025[9]. The growth would come from a line in Fukushima[9]. METI, Japan's industry ministry, chose the project for a program that supports new plants and equipment in Japan to strengthen supply chains[9]. AGC also said it was aiming for more than ¥40 billion of sales in the business by 2025[9]. Its EUV blank sales grew 50 percent in 2023, AGC said in December 2024[8]. In February 2025 AGC reported that the sales reached ¥40 billion in 2024, which it called a year ahead of its initial target[21].

4 HOYA, Information Technology business, review of operations, Integrated Report... HOYA, accessed 9 September 2026. hoya.com

7 ASML and TSMC Announce Initiative to Pioneer Industry Transition to... TSMC, 8 September 2026. pr.tsmc.com

8 The Japanese glass behind next-generation chips. AGC, 13 December 2024. agc.com

9 AGC to Boost Production Capacity of EUVL Photomask Blanks. AGC, 27 April 2023. agc.com

10 AGC Data Book. AGC Inc., 29 May 2026, p. 8. agc.com

20 5 things you should know about High NA in EUV. ASML, 25 January 2024. asml.com

21 Financial Results for FY2024. AGC Inc., 7 February 2025, p. 45. agc.com

AGC plans to add capacity again after a dip in sales. Its EUV blank sales declined in 2025 and were recovering in 2026, AGC said in June 2026[22]. In August 2026 AGC set out capacity expansions of 20 to 50 percent across EUV blanks, polishing slurry and lens materials for lithography tools[23]. It gave no separate figure for the blanks[23]. Mass production on the added capacity is to begin in 2028[23].

Writers

An electron-beam writer draws the pattern into the resist on the blank. Multi-beam writers fire many beams at once, and Japan's NuFlare sells them for the 3 nm node[19].

In September 2023 Intel, the majority owner of Austria's IMS Nanofabrication, called it the established industry leader in the multi-beam mask writing tools that advanced EUV requires[24]. Counting all electron-beam mask writers, CSET, a research center at Georgetown University, put IMS's share of a $994 million world market at 46.6 percent in 2025, using TechInsights data[3]. NuFlare had 39.2 percent and JEOL 14.2 percent, in the same CSET analysis of TechInsights data[3].

Intel sold a combined 32 percent of IMS in 2023 and still held the other 68 percent on 27 December 2025[25]. In June 2023 it agreed to sell about 20 percent of IMS to Bain Capital[24]. Intel agreed in September 2023 to sell roughly 10 percent of IMS to TSMC at a valuation of about $4.3 billion[24]. TSMC's subsidiary TSMC Development held that 10 percent at the end of 2024, at a fair value of $432.8 million in TSMC's accounts[26].

Inspection and repair

A mask defect cuts the number of good chips on every wafer that mask prints, so masks are checked as blanks, after patterning and again at the fab. Actinic inspection uses 13.5 nm light, the wavelength of EUV lithography[11]. In 2019 Lasertec launched the ACTIS A150, which it calls the world's first actinic inspection system for patterned EUV masks[11]. In October 2025 it launched the ACTIS A200HiT series,

3 Supply Chain Explorer: Advanced Chips. Emerging Technology Observatory (ETO), accessed 9 September 2026. chipexplorer.eto.tech

11 Lasertec Releases ACTIS A200HiT Series Actinic EUV Patterned Mask Inspection... Lasertec, 31 October 2025. lasertec.co.jp

19 EB Mask Writer. NuFlare Technology, accessed 9 September 2026. nuflare.co.jp

22 Semiconductor-related Business Briefing: Event Summary. AGC Inc., 2 June 2026, p. 17. agc.com

23 AGC Company Overview. AGC Inc., August 2026, p. 52. agc.com

24 Intel to Sell Minority Stake in IMS Nanofabrication Business to TSMC. Intel, 12 September 2023. intc.com

25 INTEL CORP, Form 10-K annual report for the period ended 2025-12-27 (10-K). U.S. Securities and Exchange Commission, 23 January 2026. sec.gov

26 TAIWAN SEMICONDUCTOR MANUFACTURING CO LTD, Form 6-K report of foreign private... U.S. Securities and Exchange Commission, 27 February 2025. sec.gov

which it says triples the A150's inspection speed[11]. Lasertec reported net sales of ¥230,485 million in the year to June 2026[27].

As of 2026 KLA's Teron systems for mask shops detect pattern and particle defects on both EUV masks and optical masks, the quartz masks used with deep ultraviolet light[15].

Zeiss makes AIMS EUV, which checks a mask in EUV light under the scanner's own imaging conditions to judge how a defect will affect printing[13]. Zeiss calls AIMS the only system that tests how a mask will print under conditions equivalent to the scanner's[13]. Zeiss says the tool can be extended to ASML's High-NA scanner, the EXE:5000[13]. Zeiss also sells electron-beam repair tools that work on EUV masks[28].

Pellicles

A pellicle is a thin membrane on a frame above the mask. Particles land on it, too far out of focus to print, and the pattern below stays clean.

Every photon a pellicle absorbs is light that never reaches the wafer[29]. The light crosses the pellicle twice, on the way in and on the way back off the mirror, so absorption counts double. In 2022 ASML described its most advanced EUV pellicle as a membrane 13 nanometers thick that can withstand up to 500 degrees Celsius in a high vacuum[29]. ASML released its first EUV pellicles in 2016 and calls the pellicle one of the breakthroughs that brought EUV lithography to high-volume manufacturing[29].

Carbon nanotube membranes are a newer kind of pellicle, aimed at next-generation High-NA, high-output EUV scanners[30]. In 2020 imec, a Belgian chip research center, mounted them on masks and exposed them in an ASML NXE:3300 scanner[31]. The membranes passed up to 97 percent of the light in a single crossing, and imec found their effect on imaging low and correctable[31].

Mitsui Chemicals began commercial production of silicon-based EUV pellicles under an ASML license in 2021[30]. In 2024 it said it would build a carbon nanotube membrane plant at Iwakuni-Ohtake, tentatively due to finish in December 2025[30]. In December 2025 Mitsui Chemicals said the plant was scheduled for completion in its fiscal

11 Lasertec Releases ACTIS A200HiT Series Actinic EUV Patterned Mask Inspection... Lasertec, 31 October 2025. lasertec.co.jp

13 Mask Qualification Solutions by ZEISS SMT. Carl Zeiss, accessed 9 September 2026. zeiss.com

15 Reticle Qualification. KLA Corporation, accessed 20 October 2026. kla.com

27 Lasertec, business report. Lasertec, September 2026. lasertec.co.jp

28 Mask Repair Solutions by ZEISS SMT. Carl Zeiss, accessed 9 September 2026. zeiss.com

29 Indistinguishable from magic: the EUV pellicle. ASML, 14 September 2022. asml.com

30 Mitsui Chemicals Sets up Production Facilities for CNT Pellicles to Be Used in... Mitsui Chemicals, 28 May 2024. jp.mitsuichemicals.com

31 Imec demonstrates CNT pellicle utilization on EUV scanner. Imec, 6 October 2020. imec-int.com

2025, in line with its initial plan[32]. The same presentation lists the plant under March 2026 in its table of key investments[33]. The membranes are designed to pass 92 percent or more of the light and to withstand exposure power above 1 kW[30].

03

Cost and design economics

A mask set costs more at each newer node, by SemiAnalysis's figures. In July 2022 SemiAnalysis put mask sets above $1 million at 28 nm and above $10 million at 7 nm[6]. It expected 3 nm mask sets to reach the $40 million range[6]. Its May 2026 primer still puts a leading-edge mask set at $40 million[5]. The same primer puts a single respin, remaking a chip after a design flaw, at $50 million to $100 million at leading-edge nodes[5].

Mask costs restrict which firms can use a node. SemiAnalysis wrote that at 5 nm and 3 nm, only firms with large wafer volume see the cost per transistor fall[6]. It wrote that more companies will lack the volume to spread the fixed cost of mask sets[6]. It expected failed chip designs to show up first among AI chip startups, which work on much tighter budgets[6].

Figure 05.2

In 2022 SemiAnalysis projected a 3 nm mask set at $40 million

What one mask set costs, by node, 2022. The 28 nm and 7 nm figures are minimums and 3 nm was a projection



Source SemiAnalysis, The Dark Side of the Semiconductor Design Renaissance, July 2022. SemiAnalysis's May 2026 EDA Market Primer puts a leading-edge mask set at $40M [6]

04

Who makes it

Mask making splits between chipmakers that make masks for their own use and independent merchant firms that sell to them and

5 EDA Market Primer. SemiAnalysis, 21 May 2026. newsletter.semianalysis.com

6 The Dark Side Of The Semiconductor Design Renaissance – Fixed Costs Soaring Due... SemiAnalysis, 24 July 2022. newsletter.semianalysis.com

30 Mitsui Chemicals Sets up Production Facilities for CNT Pellicles to Be Used in... Mitsui Chemicals, 28 May 2024. jp.mitsuichemicals.com

32 Business Strategy Presentation (L&HC, Mobility, ICT) in December 2025. Mitsui Chemicals, 17 December 2025, p. 31. jp.mitsuichemicals.com

33 Business Strategy Presentation (L&HC, Mobility, ICT) in December 2025. Mitsui Chemicals, 17 December 2025, p. 28. jp.mitsuichemicals.com

others[1]. Photronics says that chipmakers' own shops once supplied most of the market, and that production has more recently tended to move toward the independents[1]. TSMC makes masks too, and lists mask making among the services it offers besides making wafers[34].

Photronics calls itself a leading worldwide maker of masks for chips and flat panel displays[12]. It had $849.3 million of revenue in fiscal 2025, $615.1 million of it from chip masks[12]. The competitors it names in chip and display masks include Dai Nippon Printing, Hoya, Tekscend Photomask and two Chinese firms, Shenzhen Newway Photomask and Shenzhen Qingyi Photomask[1]. It also competes with chipmakers' own mask shops[1].

Chinese and foreign firms make masks in China, but no Chinese firm appears in CSET's 2025 market shares for electron-beam mask writers, the machines that draw the pattern onto a mask[3]. SMIC, a Chinese foundry, runs two in-house mask shops in Shanghai that make masks for deep ultraviolet and older lithography machines[14]. Photronics began operating two mask plants in China, in Hefei and Xiamen, in 2019[1]. Its chip mask business in Xiamen is a joint venture in which Dai Nippon Printing holds 49.99 percent[1].

05

The chokepoint

Every EUV mask depends on a handful of firms at each step, from the blank plate to the tool that tests how it will print. Only three firms supply the blank plates for EUV masks[8]. IMS Nanofabrication held 46.6 percent of the 2025 market for electron-beam mask writers and NuFlare 39.2 percent, by CSET's count[3]. In 2019 Lasertec launched what it calls the first tool that inspects patterned EUV masks in EUV light[11]. Zeiss calls its AIMS the only system that tests how a mask will print under conditions equivalent to the scanner's[13].

Pellicles already work in high-volume EUV manufacturing[29]. But ASML says no material is completely transparent to EUV light, so any pellicle in the light's path costs customers productivity[29]. In 2024 Mitsui Chemicals said it would set up production of carbon nanotube

1 PHOTRONICS INC, Form 10-K annual report for the period ended 2025-10-31 (10-K). U.S. Securities and Exchange Commission, 17 December 2025. sec.gov

3 Supply Chain Explorer: Advanced Chips. Emerging Technology Observatory (ETO), accessed 9 September 2026. chipexplorer.eto.tech

8 The Japanese glass behind next-generation chips. AGC, 13 December 2024. agc.com

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12 Photronics Reports Full Year and Fourth Quarter Fiscal 2025 Results. GlobeNewswire (release by Photronics), 10 December 2025. globenewswire.com

13 Mask Qualification Solutions by ZEISS SMT. Carl Zeiss, accessed 9 September 2026. zeiss.com

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29 Indistinguishable from magic: the EUV pellicle. ASML, 14 September 2022. asml.com

34 TAIWAN SEMICONDUCTOR MANUFACTURING CO LTD, Form 20-F annual report for the... U.S. Securities and Exchange Commission, 16 April 2026. sec.gov

membranes, designed to pass 92 percent or more of the light, for next-generation High-NA, high-output EUV scanners[30].

Four questions to answer from memory

1. What is a photomask? see How it works
2. How many firms make the blank plates for EUV masks? see Blanks
3. Why is it hard for a new firm to start making EUV mask blanks? see The chokepoint
4. What kind of masks does China's SMIC make in its own shops? see Who makes it

Four things to remember

- A photomask is the stencil for one layer of a chip, and any flaw on it repeats on every wafer.
- Three firms make EUV mask blanks, and Lasertec launched the first tool that checks finished masks in EUV light.
- AGC spent more than a decade learning to make EUV blanks before mass production began in 2017.
- China makes masks for deep ultraviolet machines and older ones, in SMIC's own mask shops and at independent firms such as Shenzhen Qingyi and Shenzhen Newway.

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Made by **Yashvardhan Sharma** with Claude. Corrections and questions: contact@chipsupplychain.org.

A company filings, governments, standards bodies, universities and research institutions; B SemiAnalysis, Reuters, the Financial Times and Nikkei Asia.

How to cite this chapter

> Sharma, Yashvardhan. “Photomasks and Pellicles.” *Chip Supply Chain: How AI Chips Are Made*. Edition of 27 October 2026. https://chipsupplychain.org/photomasks-and-pellicles/

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@misc{chipsupplychain2026photomasksandpellicles,
  title        = {Photomasks and Pellicles},
  author       = {Sharma, Yashvardhan},
  howpublished = {Chip Supply Chain: How AI Chips Are Made},
  year         = {2026},
  month        = {October},
  note         = {Edition of 27 October 2026},
  url          = {https://chipsupplychain.org/photomasks-and-pellicles/}
}
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